

## Ch. 12 — P01 (C) Verification and Validation Concepts

Problem Bank | Chapter 12 | Type: C | Difficulty:

---

Chapter: 12 Type: C Difficulty:

---

Answer the following questions in complete sentences. Each answer should be 2–4 sentences.

### Tasks

---

1. **Verification vs. Validation:** Define model verification and model validation. Give one example of a verification failure and one example of a validation failure that could occur in a hospital emergency department simulation.
2. **The V&V distinction in practice:** A modeler builds a SimPy queue simulation. She runs it for 10,000 time units and finds that the simulated mean wait is 4.7 minutes, while the theoretical M/M/1 formula gives 4.2 minutes. Is this a verification problem or a validation problem? What additional information would you need to decide?
3. **Extreme-condition tests:** List three extreme-condition tests that any queue simulation should pass. For each test, state:
  - (a) What condition you set
  - (b) What output you expect
  - (c) What a failure of this test tells you
4. **Face validity:** What is face validity? Who performs a face validity check, and why can't the modeler perform it alone?
5. **When is a model valid enough?** A simulation model of a hospital ward predicts average length of stay (LOS) within 5% of historical data for 8 of 10 patient types, but overestimates LOS by 25% for one rare but high-cost type. Is this model valid enough for:
  - (a) Planning routine staffing levels?
  - (b) Designing a new isolation protocol for the rare patient type?

Justify your answers.

6. **Documentation requirement:** Why must model assumptions be documented even when they are obviously true? Give an example where an “obvious” assumption turned out to be false when the model was applied to a different facility.